

# Scanning Button Email Testing

## 1. Artefact Overview

This technical artefact documents the implemented phishing-email detection prototype and demonstrates its operation using selected records from the held-out test dataset. The evidence supplied for this artefact consists of twelve end-to-end test screenshots captured from the interface. The screenshots show six phishing-labelled and six safe-labelled examples.

The evaluation interface reports that the held out test dataset contains 1,752 rows and that 50 emails were selected for the controlled interface: 25 Safe and 25 Phishing, using the fixed seed 2026. The twelve screenshots in this artefact are a qualitative subset of that 50-email interface and are used to demonstrate functional behaviour rather than to replace the project's quantitative model evaluation.

### 1.1 Artefact Objectives

- Demonstrate that a held-out dataset email can be selected and rendered in a controlled mail-client interface.
- Demonstrate end-to-end classification into Safe Email or Potential Phishing Email.
- Show confidence and class probabilities returned to the interface.
- Show supplementary phishing indicators where they are triggered.
- Demonstrate integrated SHAP explainability through factors supporting and working against prediction.
- Provide reproducible visual evidence suitable for the project demonstration and final report.

### 1.2 Scope and Evidence Boundary

The screenshots demonstrate functional and qualitative system testing. They do not, by themselves, establish aggregate accuracy, precision, recall or F1-score. The ground-truth label visible in the controlled interface is retained for evaluator comparison; it is not treated as model input.

## 2. Implemented System

The prototype combines a controlled email interface, a browser extension, a local FastAPI inference service, a DeBERTa-v3-large based classifier with engineered phishing features and SHAP-based explanation. The browser-facing component presents the classification result and explanation while the backend performs model inference.

| Stage                       | Technical role                                                                                         |
|-----------------------------|--------------------------------------------------------------------------------------------------------|
| 1. Held-out email selection | A dataset record is selected from the fixed evaluation subset.                                         |
| 2. Email rendering          | Synthetic sender/subject metadata and the unchanged dataset email text are displayed in the interface. |
| 3. Browser extraction       | The extension reads the currently displayed sender, subject and body.                                  |
| 4. API request              | The extracted email is submitted to the local FastAPI prediction                                       |

|                             |                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------|
|                             | endpoint.                                                                                                |
| 5. Classification           | The DeBERTa-v3-large gated-hybrid classifier produces Safe/Phishing probabilities and a verdict.         |
| 6. Supplementary indicators | Rule-based indicators provide additional contextual warnings without replacing the model verdict.        |
| 7. Explainability           | The explanation function returns SHAP token contributions supporting and working against the prediction. |
| 8. User presentation        | The extension renders the verdict, confidence, probabilities, indicators and explanation.                |

## 2.1 Controlled Test Design

The interface deliberately separates evaluation metadata from email content. Each selected message is assigned a synthetic sender and subject for display, while the dataset email text is retained. Links are shown as inert text in the controlled environment. The fixed selection seed supports repeatability.

## 3. Test Evidence Summary

| Case  | Interface item | Ground truth   | Observed verdict         | Functional result |
|-------|----------------|----------------|--------------------------|-------------------|
| DT-01 | Test 1         | Phishing Email | Potential Phishing Email | Pass              |
| DT-02 | Test 2         | Safe Email     | Safe Email               | Pass              |
| DT-03 | Test 3         | Phishing Email | Potential Phishing Email | Pass              |
| DT-04 | Test 4         | Safe Email     | Safe Email               | Pass              |
| DT-05 | Test 5         | Phishing Email | Potential Phishing Email | Pass              |
| DT-06 | Test 6         | Safe Email     | Safe Email               | Pass              |
| DT-07 | Test 7         | Phishing Email | Potential Phishing Email | Pass              |
| DT-08 | Test 8         | Safe Email     | Safe Email               | Pass              |
| DT-09 | Test 9         | Phishing Email | Potential Phishing Email | Pass              |
| DT-10 | Test 10        | Safe Email     | Safe Email               | Pass              |
| DT-11 | Test 11        | Phishing Email | Potential Phishing Email | Pass              |
| DT-12 | Test 12        | Safe Email     | Safe Email               | Pass              |

Across the twelve supplied screenshots, the displayed verdict agrees with the visible ground truth in every illustrated case. This is a qualitative observation about the supplied examples only and should not be interpreted as a 100% accuracy claim for the complete held-out dataset.

## 4. Detailed End-to-End Test Cases

For each case, the screenshot is treated as implementation evidence.

### 4.1 Test Case DT-01 — Test 1

| Field                | Evidence                                                     |
|----------------------|--------------------------------------------------------------|
| Purpose              | Phishing classification with explicit call-to-action content |
| Dataset ground truth | Phishing Email                                               |

|                         |                                                                                                        |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Observed system verdict | Potential Phishing Email                                                                               |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

The selected message is labelled Phishing Email in the controlled interface. The extension returns Potential Phishing Email with very high phishing probability. A suspicious call-to-action indicator is displayed, and the SHAP panel lists tokens that support and oppose the phishing prediction.

The screenshot displays a web-based email interface. On the left, a sidebar shows navigation options like 'Compose', 'Inbox' (50), 'Starred', 'Snoozed', 'Sent', and 'Drafts'. Below this is an 'EVALUATION' section with a 'Held-out test set' and 'Ground truth retained'. The main area shows an inbox of 50 messages, with 'Test 1' selected. The email content is a 'Security Notification 1' from 'security-alert-1@example.test'. A 'Ground truth: Phishing Email' label is visible. A 'Scan Email for Phishing' button is in the top right. On the far right, a panel titled 'Potential Phishing Email' shows a confidence of 99.97%, detected indicators, and a SHAP explanation for the prediction.

Figure 1. Dataset email Test 1: end-to-end classification and explainability evidence.

## 4.2 Test Case DT-02 — Test 2

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Safe classification of long-form legitimate text                                                       |
| Dataset ground truth    | Safe Email                                                                                             |
| Observed system verdict | Safe Email                                                                                             |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS - observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

The selected message is labelled Safe Email. The extension returns Safe Email with very high safe probability. The long body demonstrates that the interface can render substantial text while the explanation panel remains available.

The screenshot displays a web browser at localhost:5500/dataset\_test.html. The main interface shows an email client with an 'Inbox' of 50 messages. The selected email, 'University Services Update 2', is from 'student-services-2@example.test'. The email body contains a 'Security Notification 1' and a quote from 'Jon O. writes:'. On the right, a 'Safe Email' panel indicates a confidence of 99.98% and a SHAP explanation for the prediction. The explanation lists factors supporting the prediction, such as 'J' and 'I', and factors working against it, such as 's' and 't'.

**Figure 2. Dataset email Test 2: end-to-end classification and explainability evidence.**

### 4.3 Test Case DT-03 — Test 3

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Phishing classification of a short lure                                                                |
| Dataset ground truth    | Phishing Email                                                                                         |
| Observed system verdict | Potential Phishing Email                                                                               |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

The selected phishing-labelled record contains a comparatively short message. The system still returns Potential Phishing Email, demonstrating that classification is not dependent on a long body or a fixed template.

The screenshot displays a web-based email interface. The left sidebar shows navigation options like 'Compose', 'Inbox' (50 messages), 'Starred', 'Snoozed', 'Sent', and 'Drafts'. The main area shows an inbox list with 50 messages. The selected email is 'Security Notification 3' from 'security-alert-3@example.test'. The email body contains a phishing lure: 're: vallium viagra calls hello, would you like to spend less on your medications? visit medications - by - mail shop and save over 75 % with us. va u ag c is 8 mvl ra lal xandmany... surprised with our prices!'. A 'Ground truth: Phishing Email' label is visible below the email header. On the right, a 'Potential Phishing Email' alert is shown with a confidence of 99.97%. Below the alert, an 'Explainable AI Analysis' section provides a SHAP explanation for the prediction, listing factors supporting and working against the prediction.

**Potential Phishing Email**

Confidence: 99.97%  
Safe: 0.03% | Phishing: 99.97%

Sender: security-alert-3@example.test  
Subject: Security Notification 3

**Explainable AI Analysis**  
Identify the words and tokens that influenced the model's prediction.

**Explain Decision**

SHAP explanation generated successfully.

**Explanation for: Phishing Email**

**Factors Supporting the Prediction**

- 1. Notification → Contribution: 0.00955
- 2. 3 → Contribution: 0.00116
- 3. val → Contribution: 0.00082
- 4. → Contribution: 0.000638
- 5. re → Contribution: 0.00452
- 6. va → Contribution: 0.00194
- 7. a → Contribution: 0.00164
- 8. visit → Contribution: 0.00151
- 9. → Contribution: 0.000138
- 10. me → Contribution: 0.000137

**Factors Working Against the Prediction**

- 1. Security → Contribution: -0.007809
- 2. → Contribution: -0.001308

SHAP contributions show how individual words and tokens influenced the model's prediction.

Powered by DeBERTa-v3-large + SHAP Explainable AI

**Figure 3. Dataset email Test 3: end-to-end classification and explainability evidence.**

#### 4.4 Test Case DT-04 — Test 4

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Safe classification of a long discussion-style message                                                 |
| Dataset ground truth    | Safe Email                                                                                             |
| Observed system verdict | Safe Email                                                                                             |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

A long, prose-heavy safe-labelled message is classified as Safe Email. This provides a contrast with the phishing examples and demonstrates both output states through the same interface.

The screenshot displays a web application for email management. The left sidebar contains navigation links: Compose, Inbox (50), Starred, Snoozed, Sent, Drafts, and an EVALUATION section. The main content area shows an email titled 'University Services Update 4' from 'student-services-4@example.test'. A green label 'Ground truth: Safe Email' is present. A 'Scan Email for Phishing' button is in the top right. A 'Safe Email' modal is open on the right, showing a confidence of 99.99% and a SHAP explanation. The explanation lists factors supporting the prediction, such as '1. a → Contribution: 0.051252' and '2. to → Contribution: 0.046491', and factors working against it, such as '7. meeting → Contribution: 0.037254' and '8. j → Contribution: 0.039993'.

Figure 4. Dataset email Test 4: end-to-end classification and explainability evidence.

## 4.5 Test Case DT-05 — Test 5

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Phishing classification of financial/debt-themed content                                               |
| Dataset ground truth    | Phishing Email                                                                                         |
| Observed system verdict | Potential Phishing Email                                                                               |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

The phishing-labelled message contains financial/debt-style promotional content. The detector returns Potential Phishing Email and exposes its explanation panel for inspection.

The screenshot displays a webmail interface with a sidebar on the left containing navigation links like 'Compose', 'Inbox', 'Starred', 'Snoozed', 'Sent', and 'Drafts'. The main area shows an email titled 'Security Notification 5' from 'security-alert-5@example.test'. The email content includes a 'Ground truth: Phishing Email' label and a 'Free Debt Consolidation Information' section. On the right, a 'Potential Phishing Email' warning box is visible, showing a confidence score of 99.97% and detected indicators. Below this, an 'Explainable AI Analysis' panel provides SHAP explanation for the prediction, listing factors supporting and working against the prediction.

**Figure 5. Dataset email Test 5: end-to-end classification and explainability evidence.**

## 4.6 Test Case DT-06 — Test 6

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Safe classification of technical/discussion content                                                    |
| Dataset ground truth    | Safe Email                                                                                             |
| Observed system verdict | Safe Email                                                                                             |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

The safe-labelled record is returned as Safe Email. The screenshot demonstrates correct routing of a different legitimate content style through the same end-to-end workflow.

The screenshot displays a web application interface for email management. On the left, a sidebar shows navigation options: 'Compose', 'Inbox' (50 messages), 'Starred', 'Snoozed', 'Sent', 'Drafts', and an 'EVALUATION' section. The 'Inbox' list shows various email entries, with 'Test 6' selected. The main content area shows the details of 'Test 6', including the sender 'student-services-6@example.test', the subject 'University Services Update 6', and the body text. A 'Ground truth: Safe Email' label is visible. On the right, a 'Safe Email' overlay provides a confidence score of 99.99%, a SHAP explainability analysis, and a list of factors supporting the prediction.

**Safe Email**  
Confidence: 99.99%  
Safe: 99.99% | Phishing: 0.01%

Sender: student-services-6@example.test  
Subject: University Services Update 6

**Explainable AI Analysis**  
Identify the words and tokens that influenced the model's prediction.  
[Explain Decision]

SHAP explanation generated successfully.

**Explanation for: Safe Email**

**Factors Supporting the Prediction**

- University → Contribution: 0.151105
- 6 → Contribution: 0.131774
- Update → Contribution: 0.060507
- s → Contribution: 0.062729
- know → Contribution: 0.076687
- Contribution: 0.078880
- since → Contribution: 0.072942
- input → Contribution: 0.062311
- best → Contribution: 0.053891
- Services → Contribution: 0.040708

**Factors Working Against the Prediction**

- will → Contribution: -0.01661
- all → Contribution: -0.002914
- Contribution: -0.006427
- re → Contribution: -0.003316
- what → Contribution: -0.001757
- they → Contribution: -0.001757
- Contribution: -0.001757
- re → Contribution: -0.001757
- do → Contribution: -0.001425
- is → Contribution: -0.001425

SHAP contributions show how individual words and tokens influenced the model's prediction.

Powered by Databricks x SHAP Explainable AI

**Figure 6. Dataset email Test 6: end-to-end classification and explainability evidence.**

## 4.7 Test Case DT-07 — Test 7

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Phishing classification of promotional/enticement content                                              |
| Dataset ground truth    | Phishing Email                                                                                         |
| Observed system verdict | Potential Phishing Email                                                                               |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

The ground truth is Phishing Email and the detector returns Potential Phishing Email. Supplementary indicators and SHAP evidence are visible alongside the verdict.

The screenshot displays a web application interface for evaluating phishing emails. On the left, a sidebar shows a list of emails in the 'Inbox' (50 messages). The main area displays a selected email titled 'Security Notification 7' from 'security-alert-7@example.test'. The email content is a phishing attempt, mentioning a vacation adventure to Florida. A 'Scan Email for Phishing' button is visible in the top right corner. On the right side, a 'Potential Phishing Email' overlay is shown, providing detailed analysis:

- Confidence:** 99.97%
- Safe:** 0.03% / **Phishing:** 99.97%
- Detected Indicators:**
  - Suspicious call-to-action language detected
  - Unexpected price, reward or promotional offer detected
- Sender:** security-alert-7@example.test
- Subject:** Security Notification 7
- Explainable AI Analysis:**
  - Identify the words and tokens that influenced the model's prediction.
  - Explain Decision** button
  - SHAP explanation generated successfully.
- Explanation for: Phishing Email**
  - Factors Supporting the Prediction:**
    - 7 → Contribution: 0.039654
    - Notification → Contribution: 0.018446
    - adventure → Contribution: 0.014104
    - lucky → Contribution: 0.012045
    - for → Contribution: 0.004881
    - selected → Contribution: 0.003212
    - been → Contribution: 0.003108
    - to → Contribution: 0.002170
    - for → Contribution: 0.002167
    - ve → Contribution: 0.000868
  - Factors Working Against the Prediction:**
    - Security → Contribution: -0.025728
    - 1 → Contribution: -0.014887
    - 1 → Contribution: -0.009603
    - 1 → Contribution: -0.007848
    - you → Contribution: -0.007109
    - florida → Contribution: -0.007479
    - congratulations → Contribution: -0.007390
    - vacation → Contribution: -0.005045
    - 1 → Contribution: -0.004371

**Figure 7. Dataset email Test 7: end-to-end classification and explainability evidence.**

## 4.8 Test Case DT-08 — Test 8

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Safe classification of a short URL-containing record                                                   |
| Dataset ground truth    | Safe Email                                                                                             |
| Observed system verdict | Safe Email                                                                                             |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

The selected record is labelled Safe Email and is classified Safe. This is useful qualitative evidence that the mere presence of URL-like text does not force the interface to display a phishing verdict.

The screenshot displays a web application interface for evaluating email classification. On the left, a sidebar shows navigation options like 'Compose', 'Inbox' (50 messages), 'Starred', 'Snoozed', 'Sent', and 'Drafts'. Below this, an 'EVALUATION' section provides summary statistics: 'Held-out test set', 'Ground truth retained: 25 Safe + 25 Phishing', and 'Links shown as inert text'. The main area shows an 'Inbox' with a list of 13 test emails. The selected email, 'Test 8', is from 'student-services-8@example.test' with the subject 'University Services Update 8'. The email body contains a URL and a date. Below the email content, a status message indicates 'Held-out test dataset loaded successfully: 1752 rows available, 50 test emails selected (25 Safe + 25 Phishing, seed 2026)'. On the right, a 'Safe Email' modal window displays the classification result with a confidence of 99.98% and a phishing score of 0.02%. It also includes an 'Explainable AI Analysis' section with 'Factors Supporting the Prediction' and 'Factors Working Against the Prediction', each with a list of contributing words and their SHAP values. The modal is powered by 'DeBERTa-v3-large + SHAP Explainable AI'.

**Figure 8. Dataset email Test 8: end-to-end classification and explainability evidence.**

## 4.9 Test Case DT-09 — Test 9

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Phishing classification of motivational/marketing-style content                                        |
| Dataset ground truth    | Phishing Email                                                                                         |
| Observed system verdict | Potential Phishing Email                                                                               |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

The phishing-labelled sample is returned as Potential Phishing Email. The example broadens the qualitative evidence beyond credential-themed phishing and demonstrates behaviour on persuasive marketing-style language.

The screenshot displays a web application interface for evaluating phishing emails. On the left, a sidebar shows a 'Mail' section with a search bar and a list of 50 messages. The selected message is 'Security Notification 9' from 'security-alert-9@example.test'. The main content area shows the email body, which is a persuasive message about a training program. On the right, a 'Potential Phishing Email' alert is displayed, showing a confidence of 99.97% and a SHAP explanation for the prediction. The explanation lists factors supporting the prediction, such as 'Notification' and 'You', and factors working against it, such as 'Security'.

**Figure 9. Dataset email Test 9: end-to-end classification and explainability evidence.**

## 4.10 Test Case DT-10 — Test 10

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Safe classification of long technical text                                                             |
| Dataset ground truth    | Safe Email                                                                                             |
| Observed system verdict | Safe Email                                                                                             |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

A lengthy safe-labelled email is classified as Safe Email. The screenshot demonstrates the reader pane, classification card and SHAP explanation operating together on a large message.

The screenshot displays a webmail interface with a sidebar on the left containing navigation options like 'Compose', 'Inbox', 'Starred', 'Snoozed', 'Sent', and 'Drafts'. The main area shows an email titled 'University Services Update 10' from 'student-services-10@example.test'. The email content discusses power market developments and electricity prices. To the right of the email, a 'Safe Email' classification card is visible, showing a confidence of 99.99% and a safe score of 99.99%. Below the classification card, a SHAP explanation is provided, listing factors supporting the prediction and factors working against it. The explanation includes a list of 10 factors with their respective contributions to the model's prediction.

**Safe Email**  
Confidence: 99.99%  
Safe: 99.99% | Phishing: 0.01%

Sender: student-services-10@example.test  
Subject: University Services Update 10

**Explainable AI Analysis**  
Identify the words and tokens that influenced the model's prediction.  
Explain Decision  
SHAP explanation generated successfully.

**Explanation for: Safe Email**

**Factors Supporting the Prediction**

1. the → Contribution: 0.094576
2. the → Contribution: 0.000640
3. we → Contribution: 0.075817
4. them → Contribution: 0.067640
5. University → Contribution: 0.047284
6. Services → Contribution: 0.047079
7. developments → Contribution: 0.048010
8. energy → Contribution: 0.038439
9. industry → Contribution: 0.038439
10. → Contribution: 0.035914

**Factors Working Against the Prediction**

1. high → Contribution: -0.029028
2. underlying → Contribution: -0.013734
3. the → Contribution: -0.010844
4. factors → Contribution: -0.012816
5. major → Contribution: -0.009937
6. inform → Contribution: -0.001493
7. abstract → Contribution: -0.001493
8. ( → Contribution: -0.001493
9. some → Contribution: -0.000965
10. of → Contribution: -0.000965

SHAP contributions show how individual words and tokens influenced the model's prediction.  
Powered by ChExBERT-v2-large + SHAP Explainable AI

**Figure 10. Dataset email Test 10: end-to-end classification and explainability evidence.**

## 4.11 Test Case DT-11 — Test 11

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Phishing classification of report/offer-style content                                                  |
| Dataset ground truth    | Phishing Email                                                                                         |
| Observed system verdict | Potential Phishing Email                                                                               |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

The selected message is labelled Phishing Email and the detector returns Potential Phishing Email. The result panel again exposes probability, indicators and token-level explanation.

The screenshot displays a web application interface for email evaluation. On the left, a sidebar shows navigation options like 'Compose', 'Inbox', 'Starred', 'Snoozed', 'Sent', and 'Drafts'. The main area shows an 'Inbox' with 50 messages. A list of messages is visible, including 'security-alert-1@example.test', 'student-services-2@example.test', 'security-alert-3@example.test', 'student-services-4@example.test', 'security-alert-5@example.test', 'student-services-6@example.test', 'security-alert-7@example.test', 'student-services-8@example.test', 'security-alert-9@example.test', 'student-services-10@example.test', 'security-alert-11@example.test', 'student-services-12@example.test', and 'security-alert-13@example.test'. The selected message is 'Security Notification 11' from 'security-alert-11@example.test'. The email content is a phishing attempt titled 'SPECIAL REPORT How To Reliably Generate Hundreds Of Leads And Prospects Every Week!'. The interface also shows a 'Scan Email for Phishing' button and a detailed 'Potential Phishing Email' result panel on the right. This panel includes a confidence score of 99.97%, a safe score of 0.03%, and a phishing score of 99.97%. It lists detected indicators, such as 'Suspicious call-to-action language detected', and provides a SHAP explanation for the prediction, showing factors supporting the prediction and factors working against it.

**Figure 11. Dataset email Test 11: end-to-end classification and explainability evidence.**

## 4.12 Test Case DT-12 — Test 12

| Field                   | Evidence                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Purpose                 | Safe classification of another long-form message                                                       |
| Dataset ground truth    | Safe Email                                                                                             |
| Observed system verdict | Safe Email                                                                                             |
| Components exercised    | Held-out dataset interface → browser extension → local API → classifier → result UI → SHAP explanation |
| Outcome                 | PASS — observed verdict agrees with the ground truth shown in the supplied screenshot.                 |

The final supplied example is labelled Safe Email and receives a Safe Email verdict, completing the alternating phishing/safe demonstration set.

The screenshot displays a web application for email evaluation. The interface is divided into several sections:

- Left Sidebar:** Contains navigation links for 'Compose', 'Inbox' (50 messages), 'Starred', 'Snoozed', 'Sent', 'Drafts', and an 'EVALUATION' section with a 'Held-out test set'.
- Center Panel:** Shows a list of emails in the 'Inbox'. The selected email is 'University Services Update 12' from 'student-services-12@example.test'. The 'Ground truth' is 'Safe Email'. The email content is displayed, mentioning 'ut resumes risk', 'sally', 'brant', and 'hunter'.
- Right Panel:** Features a 'Scan Email for Phishing' button and a 'SHAP explanation generated successfully' message. Below this is a detailed 'Explainable AI Analysis' panel.

The 'Explainable AI Analysis' panel provides a breakdown of the model's prediction:

- Safe Email Prediction:** Confidence: 99.99%, Safe: 99.99%, Phishing: 0.01%.
- Factors Supporting the Prediction:**
  - 1. an → Contribution: 0.085059
  - 2. you → Contribution: 0.059588
  - 3. of → Contribution: 0.055147
  - 4. if → Contribution: 0.053447
  - 5. which → Contribution: 0.045554
  - 6. University → Contribution: 0.045058
  - 7. 12 → Contribution: 0.043772
  - 8. our → Contribution: 0.039818
  - 9. slack → Contribution: 0.030314
  - 10. the → Contribution: 0.028532
- Factors Working Against the Prediction:**
  - 1. risk → Contribution: -0.000884
  - 2. , → Contribution: -0.000884
  - 3. is → Contribution: -0.000884

The SHAP explanation shows how individual words and tokens influenced the model's prediction. The interface is powered by DuBERTv4-large and SHAP Explainable AI.

**Figure 12. Dataset email Test 12: end-to-end classification and explainability evidence.**

## 5. Factors Working Against the Prediction and Factors Supporting the Prediction

These two sections are the SHAP explanation of the model's decision. They answer a different question from the probability score: "Which words/tokens in this particular email pushed the model toward or away from the prediction it made?"

### Factors Supporting the Prediction

These are words/tokens whose SHAP contributions push the model toward the class it predicted.

For example, suppose the result is:

Prediction: Potential Phishing Email - 98%

and the explanation shows:

Factors Supporting the Prediction

verify +0.31

password +0.24

urgent +0.18

account +0.11

This means that, for this particular email and model decision, those tokens contributed in the direction of the Phishing prediction. A larger positive SHAP magnitude generally means the token had a stronger contribution to that prediction.

Importantly, it does not mean "password always means phishing." The contribution is contextual and specific to the model and the email being explained.

### Factors Working Against the Prediction

These are tokens whose SHAP contributions push away from the predicted class, effectively providing evidence in the opposite direction.

Using the same example:

Prediction: Potential Phishing Email

Supporting:

verify +0.31

password +0.24

urgent +0.18

Working Against:

meeting -0.12

university -0.08

thanks -0.04

Conceptually, the model is receiving competing evidence.

We can think of it as:

**PHISHING**

↑

verify +0.31

password +0.24

urgent +0.18

|

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|

thanks -0.04

university-0.08

meeting -0.12

↓

**SAFE**

The phishing supporting evidence is stronger overall, so the final prediction remains Phishing.

It also works when the prediction is Safe

Suppose:

Prediction: Safe Email - 99%

Then Factors Supporting the Prediction are tokens pushing the model toward Safe, because Safe is the predicted class. Meanwhile, Factors Working Against the Prediction are tokens pushing away from Safe i.e. toward Phishing.

So the headings do not permanently mean:

- Supporting = phishing words
- Against = safe words

Instead they mean:

| Model prediction | Supporting factors   | Working-against factors               |
|------------------|----------------------|---------------------------------------|
| Phishing         | Push toward Phishing | Push away from Phishing / toward Safe |
| Safe             | Push toward Safe     | Push away from Safe / toward Phishing |

The DeBERTa gated-hybrid model makes the classification first. SHAP is then used to explain that decision.

Email



DeBERTa + engineered features



Gated-hybrid classifier



Safe / Phishing probabilities



Final prediction



SHAP analyses the model



Supporting / opposing token contributions

Therefore, clicking Explain Decision does not change the prediction or probability. SHAP explains the model output that has already been generated. The displayed SHAP values tell us the direction and relative contribution of individual tokens to the explained output. For example, a SHAP contribution of +0.25 does not mean 25% phishing probability.

## 12-feature phishing indicators

These are also different from SHAP. The system has engineered features such as urgency, credential-related terms, threat language, financial terms, CTA language, URLs, email addresses, phone numbers, punctuation, uppercase ratio and text length.

The extension can display something like: “Urgent or threatening language detected”

That is a supplementary indicator.

SHAP's:

Factors Supporting the Prediction is an explanation of model behaviour.

Thus, Factors Supporting the Prediction identify tokens with SHAP contributions that move the model output toward the class predicted for the selected email. Conversely, Factors Working Against the Prediction identify tokens whose contributions move the model output away from the predicted class. For a phishing prediction, supporting factors provide evidence in the phishing direction while opposing factors provide evidence in the safe direction; this relationship is reversed when the predicted class is Safe. The magnitude of a SHAP contribution represents the relative influence of that token on the explained model output and should not be interpreted as a probability. Importantly, SHAP is applied after inference to explain the classifier's behaviour; the displayed SHAP factors do not independently determine or modify the classification result.

## 6. Explainability Evidence

The result panel includes an Explainable AI Analysis section. After explanation generation, the interface separates token contributions into factors supporting the prediction and factors working against the prediction. This is valuable because the artefact does not expose only a class label: it also provides a trace of which token-level signals influenced the model output.

- Positive SHAP contributions indicate tokens pushing the model toward the displayed class.
- Negative SHAP contributions indicate tokens working against the displayed class.
- The explanation is supplementary evidence and should not be interpreted as a causal proof of why an email is malicious or legitimate.
- The screenshots demonstrate that explanation generation is integrated into the deployed interface, not only an offline analysis step.

## 7. Functional Acceptance Assessment

| Requirement                | Acceptance evidence                                                                     | Status |
|----------------------------|-----------------------------------------------------------------------------------------|--------|
| Load held-out dataset      | Interface reports successful loading of the held-out test set and 1,752 available rows. | Pass   |
| Balanced controlled subset | Interface reports 50 selected messages: 25 Safe and 25 Phishing.                        | Pass   |

|                                  |                                                                                |      |
|----------------------------------|--------------------------------------------------------------------------------|------|
| Render selected email            | Sender, subject, ground truth and body are visible for each demonstrated test. | Pass |
| Produce both verdict classes     | Supplied evidence includes six phishing and six safe verdicts.                 | Pass |
| Display probabilities/confidence | Classification cards show confidence and Safe/Phishing probabilities.          | Pass |
| Display supplementary indicators | Indicator area is visible and populated when relevant.                         | Pass |
| Generate explanation             | SHAP explanation success state and contribution lists are visible.             | Pass |
| Preserve evaluator ground truth  | Ground-truth badge remains visible for controlled comparison.                  | Pass |

## 8. Interpretation of the Demonstration

The supplied evidence demonstrates a functioning end-to-end prototype rather than an isolated model notebook. A user can select an evaluation email, view it in a mail-client-style interface, invoke the detector, receive a verdict and probabilities, and request an explanation. The alternating safe/phishing examples also demonstrate that the interface is capable of representing both decision outcomes.

The twelve illustrated cases all agree with their displayed ground-truth labels. This is useful functional evidence, but it is intentionally not converted into a performance percentage because the screenshots are a selected qualitative subset. Formal model effectiveness must be supported by the complete held-out evaluation and confusion-matrix results.

## 9. Limitations and Technical Considerations

- The screenshots cover 12 selected emails for the controlled interface; they are illustrative rather than exhaustive.
- The controlled interface uses synthetic sender addresses and subjects, so the evidence primarily evaluates text classification and integration rather than sender-authentication mechanisms.
- Browser-extension extraction is demonstrated in the controlled test environment; this evidence does not establish robust extraction across every live Gmail layout or future DOM change.
- The classifier operates on text and engineered indicators; attachment contents and broader mail-transport/authentication signals are outside the evidence shown here.
- SHAP explanations identify influential tokens but do not prove malicious intent.
- Long emails may be constrained by the model's configured token limit; therefore explanation and classification evidence should be interpreted within the deployed preprocessing configuration.
- False positives and false negatives remain possible and should be documented through the full evaluation rather than hidden by selecting only successful screenshots.

## 10. Reproducibility Notes

For reproducibility, the controlled page uses the held-out phishing\_test.csv source, a fixed seed of 2026, and a balanced selection of 25 Safe and 25 Phishing records. The test artefact should be executed with the local web server and FastAPI backend running, followed by the browser extension in the controlled page. Predictions should be recorded without rewriting test-email text after results are observed.

## 11. Technical Artefact Conclusion

The evidence demonstrates an integrated phishing-email detection prototype that connects held-out email data, a Gmail-style evaluation interface, browser-based scanning, model inference, supplementary indicators and SHAP explainability. The supplied test cases show successful functional processing of both phishing and safe messages. Together with the project's quantitative evaluation, API implementation and model-development evidence, this artefact provides a clear demonstration that the proposed system was implemented as an operational end-to-end prototype.

## Appendix A - Evidence Inventory

| Evidence                   | Description                                                                      |
|----------------------------|----------------------------------------------------------------------------------|
| Figures 1, 3, 5, 7, 9, 11  | Phishing-labelled examples with Potential Phishing Email verdicts.               |
| Figures 2, 4, 6, 8, 10, 12 | Safe-labelled examples with Safe Email verdicts.                                 |
| Classification panel       | Verdict, confidence, Safe/Phishing probabilities and supplementary indicators.   |
| Explainable AI panel       | SHAP contribution lists supporting and working against the displayed prediction. |